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TELECOMMUNICATIONS CONNECTOR WITH LATCH RELEASE MECHANISM

Abstract

The present disclosure relates to a telecommunications connector. The connector includes at least one connector portion including a front housing portion coupled to a rear housing portion, the front housing portion being rotatable relative to the rear housing portion about a longitudinal axis, the front housing portion including a ferrule and a latch, the latch defining front and rear ends and being pivotable relative to the rest of the front housing portion at a connection portion. A boot mounted on the telecommunications connector is movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of the front housing portion and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the rear housing portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of U.S. patent application Ser. No. 18/000,260, filed on Nov. 29, 2022; which is a National Stage Application of PCT/US2021/034612, filed on May 27, 2021; which claims the benefit of U.S. Patent Application Ser. No. 63/032,170, filed on May 29, 2020, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates generally to telecommunications systems. More particularly, the present disclosure relates to optical fiber communications systems and fiber optic connectors used in optical fiber communications systems.

BACKGROUND

[0003] Fiber optic communications systems are becoming prevalent in part because service providers want to deliver high bandwidth communication capabilities (e.g., data and voice) to customers. Fiber optic communications systems employ a network of fiber optic cables to transmit large volumes of data and voice signals over relatively long distances. Optical fiber connectors are an important part of most fiber optic communication systems. Fiber optic connectors allow two optical fibers to be quickly optically connected without requiring a splice. Fiber optic connectors can be used to optically interconnect two lengths of optical fiber. Fiber optic connectors can also be used to interconnect lengths of optical fiber to passive and active equipment.

[0004] A typical fiber optic connector includes a ferrule assembly supported at a front end of a connector housing. A spring is used to bias the ferrule assembly in a forward direction relative to the connector housing. The ferrule functions to support an end portion of at least one optical fiber (in the case of a multi-fiber ferrule, the ends of multiple fibers are supported). The ferrule has a front end face at which a polished end of the optical fiber is located. When two fiber optic connectors are interconnected, the front end faces of the ferrules abut one another and the ferrules are forced rearward relative to their respective connector housings against the bias of their respective springs. With the fiber optic connectors connected, their respective optical fibers are coaxially aligned such that the end faces of the optical fibers directly oppose one another. In this way, an optical signal can be transmitted from optical fiber to optical fiber through the aligned end faces of the optical fibers. For many fiber optic connector styles, alignment between two fiber optic connectors is provided through the use of an intermediate fiber optic adapter.

[0005] A fiber optic connector is often secured to the end of a corresponding fiber optic cable by anchoring strength members of the cable to the connector housing of the connector. Anchoring is typically accomplished through the use of conventional techniques such as crimps or adhesives. Anchoring the strength members of the cable to the connector housing is advantageous because it allows tensile load applied to the cable to be transferred from the strength members of the cable directly to the connector housing. In this way, the tensile load is not transferred to the ferrule assembly of the fiber optic connector. If the tensile load were to be applied to the ferrule assembly, such tensile load could cause the ferrule assembly to be pulled and result in optical disconnection

between the connector and its corresponding mated connector.

[0006] An example of a prior art LC style fiber optic connector **20** is illustrated in FIGS. **1** and **2**. As shown in FIGS. **1** and **2**, the conventional LC connector **20** includes a connector housing **22** defining a front housing portion **24** and a rear housing portion **26**. The LC connector **20** includes a ferrule assembly **28** defined by a ferrule **30**, a hub **32**, and a spring **34**. A rear end **36** of the ferrule **30** is secured within the ferrule hub **32**. When the LC connector **20** is assembled, the ferrule hub **32** and the spring **34** are captured between the front housing portion **24** and the rear housing portion **26** of the connector housing **22** and a front end **38** of the ferrule **30** projects outwardly beyond a front end **40** of the connector housing **22**. The spring **34** is configured to bias the ferrule **30** in a forward direction relative to the connector housing **22**. According to some versions, the front housing portion **24** may be formed from a molded plastic.

[0007] In the depicted example of the prior art LC connector **20**, a strain relief boot **56** is slid over a rear end **58** of the rear housing portion **26** and snaps over a boot flange **60** to retain the boot **56** with respect to the connector housing **22**. The rear housing portion **26** may define a crimp region **62** for crimping a fiber optic cable's strength layer to the rear housing portion **26**, normally with the use of a crimp sleeve (not shown). An exterior surface **64** of the rear housing portion **26** defining the crimp region **62** can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the crimp on the housing **22**.

[0008] Still referring to FIGS. **1** and **2**, the front housing portion **24** defines a latch **42** extending from a top wall **44** of the front housing portion **24** toward the rear end **46**, the latch **42** extending at an acute angle with respect to the top wall **44** of the front housing portion **24**. The rear housing portion **26** also includes a latch trigger **48** that extends from the rear end **46** of the front housing portion **24** toward the front end **40**. The latch trigger **48** also extends at an acute angle with respect to the top wall **44**. The latch trigger **48** is configured to come into contact with the latch **42** for flexibly moving the latch **42** downwardly.

[0009] As is known in the art, when the fiber optic connector **20** is placed in an LC format adapter **100**, an example of which is shown in FIGS. **3-5**, for optically coupling light from two optical fibers together, the latch **42** functions to lock the fiber optic connector **20** in place within the adapter **100**. The fiber optic connector **20** may be removed from the adapter **100** by depressing the latch trigger **48**, which causes the latch **42** to be pressed in a downward direction, freeing catch portions **52** of the latch **42** from the fiber optic adapter **100**.

[0010] In dense panel environments, where a large number of connectors, such as connector **20**, are placed in a side-by-side arrangement, access becomes a big issue. Since the latch triggers **48** have to be accessed and pushed down for unlatching the latches **42** from the adapters **100**, removal of connectors can become difficult and cumbersome. In dense panel designs, even though it is normally fairly easy to insert and latch connectors such as connectors **20** by pushing from an end of the connector's boot, removal is difficult because the latches are far forward on the connectors.

[0011] Improvements in connector design are desired to facilitate use in high density environments.

SUMMARY

[0012] One aspect of the present disclosure relates to a telecommunications connector (e.g., a fiber optic connector) having features that facilitate connector latching and unlatching.

[0013] Another aspect of the disclosure relates to a duplex telecommunications connector design, where front housings of two connector portions that are positioned side-by-side can be rotated about their longitudinal axes to change the polarity of the two connector portions.

[0014] According to one aspect, the disclosure is directed to a telecommunications connector, for example, a fiber optic connector, that includes at least one connector portion including a front housing portion coupled to a rear housing portion. The front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch. The latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion. According to one

embodiment, the connection portion is an intermediate connection portion positioned between the front end and the rear end of the latch.

[0015] The telecommunications connector also includes a yoke coupled to the rear housing portion of the at least one connector portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, and a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear end of the latch and cause the front end of the latch to pivot toward the ferrule of the front housing portion as the boot is moved away from the yoke.

[0016] According to another aspect of the disclosure, the telecommunications connector is a duplex fiber optic connector that includes two connector portions provided in a side-by-side relationship, each connector portion including a front housing portion coupled to a rear housing portion, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions. Each front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion. A yoke is coupled to the rear housing portion of each of the connector portions, wherein the rear housing portions are rotationally and axially fixed with respect to the yoke. A boot is mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear ends of each of the latches and cause the front ends of each of the latches to pivot toward the ferrules of the front housing portions as the boot is moved away from the yoke.

[0017] According to another aspect of the disclosure, a telecommunications connector comprises at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, a yoke coupled to the rear housing portion of the at least one connector portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, and a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage at least a portion of the latch and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the yoke.

[0018] According to yet another aspect of the disclosure, a telecommunications connector comprises a front housing portion coupled to a rear housing portion and a crimp region, wherein the front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, the crimp region configured for crimping at least a portion of a cable terminated by the ferrule, wherein the crimp region defines a plurality of spaced apart crimp rings stacked longitudinally, each crimp ring defining discretely spaced scalloped portions that are positioned peripherally around the rings, wherein the scalloped portions between adjacent rings are provided in a staggered position in a longitudinal direction defined by the telecommunications connector.

[0019] According to yet another embodiment of the disclosure, a telecommunications connector includes at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, and a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of the front housing portion and cause the latch to pivot relative to the rest of the front housing portion as the boot is moved relative to the rear housing portion.

[0020] A variety of additional aspects will be set forth in the description that follows. The aspects relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. 1 is a top view of a conventional LC style fiber optic connector.

[0022] FIG. 2 is a cross-sectional view that longitudinally bisects the fiber optic connector of FIG. 1.

[0023] FIGS. 3-4 are two perspective views of a conventional duplex adapter for mating with the fiber optic connector of the present disclosure.

[0024] FIG. 5 is a cross-sectional view of the duplex adapter of FIGS. 3-4.

[0025] FIG. 6 is a perspective view of one example fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0026] FIG. 7 is a side view of the fiber optic connector of FIG. 6, shown in the neutral position.

[0027] FIG. 8 is a side view of the fiber optic connector of FIG. 6, shown in an unlatched configuration.

[0028] FIG. 9 illustrates the rear end regions of the latches of the fiber optic connector of FIGS. 6-8.

[0029] FIG. 10 illustrates the strain relief boot portion of the fiber optic connector of FIGS. 6-8.

[0030] FIG. 11 is a cross-sectional view illustrating the interaction between the rear end of the latch of the fiber optic connector of FIG. 9 and the boot of FIG. 10.

[0031] FIG. 12 illustrates the front and rear housings of the connector portions of the fiber optic connector of FIGS. 6-8 separated from the yoke and the boot of the fiber optic connector.

[0032] FIG. 13 is a perspective view of the fiber optic connector of FIGS. 6-8 with a portion of the front housing broken away to illustrate the internal features thereof.

[0033] FIG. 14 illustrates the fiber optic connector of FIG. 13 with one of the front housings of a connector portion being rotated to reverse the polarity of that connector portion.

[0034] FIG. 15 is a rear perspective view of the yoke of the fiber optic connector of FIGS. 6-8 shown in isolation.

[0035] FIG. 16 is a partial assembly view of the fiber optic connector of FIGS. 6-8 illustrating cross-sectional views bisecting the front and rear housings of the connector portions.

[0036] FIG. 17 illustrates another partial assembly view of the fiber optic connector of FIG. 16 with a cross-sectional view of the yoke added in.

[0037] FIG. 18 illustrates the partial assembly view of the fiber optic connector of FIG. 17 with one of the connector portions coupled to the yoke.

[0038] FIG. 19 illustrates the connector portions coupled to the yoke of the fiber optic connector of FIGS. 6-8 with a crimp sleeve placed on the crimp region of the yoke.

[0039] FIG. 20 illustrates a cross-sectional view of a rear end of the strain relief boot to illustrate the external flared portion and the enlarged inner radius thereof.

[0040] FIG. 21 illustrates one example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. 7.

[0041] FIG. 22 illustrates another example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. 7.

[0042] FIG. **23** illustrates the boot-return feature of FIG. **22** in a flexed position.

[0043] FIG. **24** illustrates yet another example embodiment of a biased boot-return feature that can be implemented on the fiber optic connector for returning the boot toward the forward position of FIG. **7**.

[0044] FIG. **25** illustrates a top perspective, cross-sectional view of a portion of another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0045] FIG. **26** illustrates a bottom perspective, cross-sectional view of a portion of the fiber optic connector of FIG. **25**.

[0046] FIG. **27** illustrates a top perspective, cross-sectional view of a portion of another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0047] FIG. **28** illustrates a top perspective, cross-sectional view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0048] FIG. **29** illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0049] FIG. **30** illustrates a top perspective exploded view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure.

[0050] FIG. **31** illustrates a top view of the fiber optic connector of FIG. **30**.

[0051] FIG. **32** illustrates a top view of the connector of FIGS. **30** and **31** in an assembled configuration.

[0052] FIG. **33** illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector having features that are examples of inventive aspects in accordance with the present disclosure, the fiber optic connector having features similar to the connector of FIG. **28**.

DETAILED DESCRIPTION

[0053] Referring now to FIGS. **6-24**, an example embodiment of a connector **110** having features that are examples of inventive aspects in accordance with the present disclosure is illustrated. The illustrated connector **110** is a fiber optic connector that is used for aligning optical fibers. However, it should be noted that the inventive features discussed herein are fully applicable to other types of connectors that are not necessarily fiber optic connectors.

[0054] The connector **110** may be considered a duplex LC format connector, transitioning optical fibers from a single optical cable to two front fiber optic connector portions **112** as will be discussed in further detail below. In the illustrated embodiment, each connector portion **112** may define an LC profile, meaning that the connector portions **112** can mate to other LC type connectors via an LC adapter, such as the adapter **100** shown in FIGS. **3-5**, as noted above. As such, it should be noted that certain structures and functionality of those structures may be similar to the simplex LC connector **20** discussed above with respect to FIGS. **1-2**.

[0055] Still referring now to FIGS. **6-24**, the connector portions **112** of the illustrated connector **110** each houses a ferrule **78** for terminating the optical fiber of a fiber optic cable. As noted above, connector **110** may also be referred to as duplex connector assembly or simply a duplex connector. As noted above, connector **110** is mateable via the adapter **100** shown in FIGS. **3-5**. Adapter **100** mates two of the connectors **110** together or mates another LC format connector to connector **110** for fiber optic signal transmission.

[0056] In the depicted embodiment of the connector **110**, the two connector portions **112** are arranged side-by-side and in a parallel position, as illustrated in FIG. **6**. Each connector portion **112** is formed from a front housing **132** that is coupled to and rotatable relative to a rear housing **90**. A ferrule assembly **76** is positioned within the front housing **132**. Ferrule assembly **76** includes the

ferrule **78**, a hub **80** which holds the ferrule **78**, and a spring **82** which is positioned between a rear shoulder **81** of the ferrule hub **80** and a front end **91** of the rear housing **90**. The spring **82** is configured to bias hub **80** and ferrule **78** forwardly.

[0057] As shown in the internal details of the connector **110** in FIGS. **13** and **14**, an internal tube **84** may be provided extending from the ferrule hub **80**. Tube **84** may be configured to prevent epoxy from interfering with the movement of the ferrule **78**, the hub **80**, and the spring **82**.

[0058] As will be discussed in further detail below, both of the rear housings **90** of the two connector portions **112** that are positioned in a side-by-side arrangement are received by a yoke **96** through side slots **98** defined by the yoke **96**. The yoke is illustrated in isolation in FIG. **15**.

[0059] The rear housings **90** are rotationally fixed with respect to the yoke **96** once laterally received therein. As will be discussed in further detail below, it is the front housings **132** of the connector portions **112** that are rotatable relative to the rear housings **90** and the yoke **96** to change the polarity of the overall connector **110**, as illustrated in FIGS. **13-14**.

[0060] The yoke **96** defines a rear crimp region **104** for allowing crimping portions of a fiber optic cable via a crimp sleeve **106**, as shown in FIGS. **15** and **19**.

[0061] In an example embodiment, a fiber optic cable may include an outer jacket, strength members, typically in the form of aramid yarns, and two fibers **156**, **158** as illustrated in the present disclosure. Each fiber **156**, **158** is to be terminated to one of the two connector portions **112** of the connector **110**. Each fiber **156**, **158** may typically include an outer coating and a bare fiber. Typically, the coating is removed and the bare fiber is inserted into the ferrule **78**, and affixed, such as with epoxy. The aramid yarns are crimped to the crimp region **104** of the yoke **96** as will be discussed in further detail below.

[0062] Referring now to FIGS. **7-9**, each front housing **132** of the connector portions **112** defines a latch **118** with a front end **124** and a rear end **122**. Latch **118** is designed with a certain amount of elastic flexibility to pivot around a connection point **126** during latching and unlatching of latch **118** from an adapter such as adapter **100**. Latch **118** is the portion of the connector **110** that is designed to secure connector **110** to adapter **100**. The latch **118** functions similar to that of a conventional LC connector as shown in FIGS. **1-2** in coupling the connector **110** to the adapter **100**. The latch **118** includes a pair of shoulders **128** at the front end **124** (as shown in FIG. **6**) that mate with latching shoulders **208** of adapter **100** to secure the connector **110** to the adapter **100** (as shown in FIGS. **3-5**). As noted, in the illustrated embodiment, each connector portion **112** defines an LC profile, meaning that the connector portion **112** can mate with an LC format adapter. The rear end **122** of each latch includes an angled-down portion **123** that is configured to interact with a strain relief boot of the connector **110** as will be discussed in further detail below.

[0063] Illustrated in further detail in FIGS. **13**, **14**, and **16-18**, each front housing **132** includes an inner passage **136** for receiving the ferrule assembly **76**. The inner passage **136** includes a hub mount **137** toward the front of the front housing **132** that defines a keyed configuration for receiving the ferrule hub **80** in a predetermined keyed orientation. The exterior surface of the ferrule hub **80** is designed with a plurality of flats **85** for rotationally tuning the ferrule hub **80** and axially inserting the ferrule hub **80** within the hub mount **137** in a certain keyed orientation. The flats **85** defined on the surface of the ferrule hub **80** and mating surfaces of the hub mount **137** rotationally fix the ferrule hub **80** with respect to the front housing **132**. In this manner, when the front housing is rotated, the ferrule hub **80** rotates therewith.

[0064] The hub mount **137** defines a tapered front shoulder **139** for contacting the front of the ferrule hub **80** and limiting further forward movement of the ferrule hub **80**. As noted above, when the front housing **132** rotates, the ferrule hub **80** and the epoxied ferrule **78**, including the optical fiber **156/158** terminated thereto, rotates with the front housing **132**, all relative to the rear housing **90**, as shown in FIG. **14**.

[0065] The front housings **132** are inserted into, for mating with similar format connectors, inner passages **202** of adapter **100**. The adapter **100** includes two side-by-side passages **202** and two

ferrule alignment sleeves **204** therewithin. Each alignment sleeve **204** is configured to axially align the ferrules **78** for mating two similar connectors.

[0066] The front housings **132** can be unitarily molded with the latches **118** as a one-piece element. In the depicted example, the front housings **132** define an LC profile for mating with ports **210** of adapter **100**. While the example connector **110** is depicted as a duplex connector that includes two fiber optic connector portions **112**, it is to be appreciated that the inventive features discussed herein are fully applicable to a connector **110** that may include a single connector portion **112**.

[0067] As mentioned above, the connector **110** of the present disclosure is designed such that a strain relief boot **114** that is mounted to and slidable with respect to the yoke portion **96** of the connector allows the latching/unlatching functionality of the connector **110**. The strain relief boot is illustrated in further detail in FIGS. **10**, **11**, **13**, **14**, and **20**.

[0068] Referring to FIGS. **10**, **13**, **14**, and **20**, the boot **114** defines a throughhole **149**. The throughhole **149** forms a front opening **154** for receiving the yoke **96**. Boot **114** includes a flexible rear portion **159** to act as strain relief for the optical fibers **156**, **158** that pass through the boot **114**. The boot **114** is designed to be movable away from connector portions **112**, also relative to the yoke **96**, in a longitudinal direction (Direction A in FIG. **8**) causing pivoting movement of latch **118** about connection point **126** (Direction B in FIG. **8**). Such pivoting movement allows for unlatching of connector portions **112** from adapter **100**. Boot **114** simultaneously moves both latches **118** to allow for connector **110** to be unlatched from a duplex adapter or adapters **100** with side-by-side ports **210**.

[0069] Referring now to FIG. **20**, a flared exterior portion **115** at a rear end **117** of the flexible rear portion **159** of the boot **114** acts as a grip portion to facilitate pulling of the boot **114** rearward for unlatching the connector **110**. As illustrated in FIG. **20**, the rear end of the opening or throughhole **149** of the boot **114** also follows the general shape of the flared exterior portion **115** to provide a flared interior surface **119**. The flared, enlarged, interior surface **119** defines an inside radius that limits sharp cable bends when load is applied to the cable, to decrease bending loss.

[0070] Still referring to the boot **114** as shown in FIGS. **7-10**, **13**, **14**, and **20**, boot **114** defines a sloped exterior surface **151** adjacent the front end that is configured to interact with the angled portions **123** of the rear ends **122** of the latches **118** during slidable movement of the boot **114**. A lifting motion results for the rear ends **122**, which results in a downward movement of front ends **124** of latch **118**, when boot **114** is pulled longitudinally away from a remainder of connector **110**. Compare FIGS. **7** and **8**. A user can pull on boot **114** in a longitudinal direction away from the ferrules **78**, and remove the connector **110** from the adapter **100**, without directly engaging the latches **118**.

[0071] As shown in detail in FIGS. **9-11**, the boot **114** also includes a pair of side-by-side slots or recesses **153** which receive latch protrusions **113** defined underneath the angled portions **123** of the rear ends **122** of latches **118**. The protrusions **113** and the recesses **153** interact as stops during polarity reversal of the connector portions **112** such that the front housings **132** end up in the correct position with respect to the rear housings **90** after rotation. The exiting sides of the recesses **153** include a bevel **109** to facilitate rotation of the latch **118** out of a given recess **153** with little force. The bevels **109** help with starting rotation of the front housings **132** for polarity reversal.

[0072] The boot **114** includes similar slots **153** on an opposing side of the boot **114**. Thus, the boot **114** stays stationary during rotation of the front housings **132** for polarity reversal.

[0073] For the latching and unlatching functionality, a tab **111** defined on the yoke **96** (as shown in FIGS. **11** and **15**) mates with an aperture **155** defined on the boot **114** to allow longitudinal movement of boot **114** relative to yoke **96**. The tab **111**, however, is configured to abut a front end **107** of the aperture **155** to act as a stop for the slidable boot **114**.

[0074] Now referring generally to FIGS. **16-19**, to assemble the connector **110**, the fiber optic cable is initially inserted through boot **114** and the crimp sleeve **106**. The fibers **156**, **158** are inserted through the rear housings **90** and the tubes **84** and affixed to the ferrules **78** of the ferrule

assemblies **76**. The ferrule assemblies **76** are keyed as discussed above and placed within the front housings **132**. As shown in FIGS. **16-18**, the rear housings **90** are axially inserted and snapped into the front housings **132** with springs **82** positioned between the ferrule hubs **80** and the rear housings **90**. The tubes **84** extend rearward from the ferrule hubs **80** and are partially positioned within both the front housings **132** and the rear housings **90**.

[0075] As noted above, the rear housings **90** and the optical fibers **156**, **158** are inserted in a lateral direction through the side slots **98** of the yoke **96**, as shown in FIGS. **17-19**. The side slots **98** of the yoke **96** enable the fibers **156**, **158** to come in from the sides of the yoke **96** and allow the connector portions **112** to be epoxied and polished before the cable has to be crimped to the overall connector **110**. It should be noted that the boot **114** may define internal ribs that are configured to protrude into the side slots **98** of the yoke **96** for the purposes of constraining the fibers **156**, **158** within the side slots **98** and limit deflection of the fibers. As such, if the ferrule(s) **78** are pressed rearward under a bias and the terminated fibers **156**, **158** are pushed back inside the connector **110**, the fibers **156**, **158** are forced back into the jacket of the cable, guided by a predetermined path provided within the yoke **96**, to limit deflection or unwanted bending of the fibers.

[0076] Once the connector portions **112** have been placed on the yoke **96** with rear housings **90** having been inserted laterally into the yoke **96**, the cable jacket and strength members in the form of aramid yarns may be crimped to the crimp region **104** of the yoke **96** via the crimp sleeve **106**, as shown in FIG. **19**. As shown, the crimp region **104** is provided as a rear projection **99** that extends from the yoke **96**.

[0077] Referring generally to FIGS. **15** and **19**, an exterior surface **97** of the rear projection **99** that defines the crimp region **104** can be textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the strength members (e.g., aramid yarns) on the yoke **96**. In the depicted example, the exterior surface **97** defines a series of parallel rings or ridges **95** extending in a direction from the rear end of the rear projection **99** toward the main body of the yoke **96**. As shown, each of the rings **95** define discretely spaced scalloped portions **93** that are positioned peripherally around the rings **95**. The scalloped portions **93** define teeth **201** thereinbetween. As illustrated, the scalloped portions **93** (and the teeth **201**) are provided in an offset or staggered relationship between adjacent rings **95** as the rings **95** extend front to back. In this manner, when the strength members are pressed underneath the crimp sleeve **106** against the exterior surface **97** of the rear projection **99**, the strength members will tend to spread out and follow zig-zagging patterns as they are captured underneath the crimp sleeve **106**, leading to increased yarn and jacket retention. The staggered design of the scallops **93** and the teeth **201** provided therebetween force the strength members laterally into the scallops **93** in a zig-zagging pattern, such that the strength members do not simply lay parallel to the rear projection **99** in a front-to-back direction, but make multiple lateral bends over multiple edges, to increase the amount of pull force needed to separate the cable from the connector **110**.

[0078] As shown in FIG. **15**, the rear projection **99** may include a step down portion **89** with full annular rings **95** that are smaller in diameter for engaging with the jacket of the cable and providing jacket retention.

[0079] Once the cable is crimped to the yoke **96**, the boot **114** is pulled over the yoke **96** as the yoke enters the front opening **154** of the boot **114** until the rear ends **122** of the latches **118** are in the slots **153** of boot **114** and the tab **111** of the yoke **96** is within the aperture **155** defined on the boot **114** (as shown in FIGS. **9-11**). As noted previously, the tab **111** is configured to abut a front end of the aperture **155** to act as a stop for the boot **114** when the boot **114** is slid rearwardly.

[0080] Now referring specifically to FIGS. **12-14**, the polarity reversal functionality of the connector **110** will be discussed in further detail. As noted above, in the illustrated example of the connector **110**, the front housing **132** includes the ferrule assembly **76** that includes the ferrule hub **80**, which is keyed and rotationally fixed to the front housing **132** such that the ferrule hub **80** and thus, the ferrule **78** rotate with the front housing **132** during polarity reversal.

[0081] To switch polarity of connector portions **112**, the front housings **132** are rotated in opposite directions so that the rear ends **122** of the latches **118** are moved from slots **153** on a first side of the boot **114** to slots **153** positioned on the opposite side of the boot **114**. During polarity switching, boot **114** remains coupled to the yoke **96**.

[0082] During the polarity switch, the front housing **132** slidably rotates over the rear housing **90** to change the polarity of the overall connector **110**.

[0083] The rear housing **90** forms a front annular recess or notch **250** that is defined between a front annular ring **251** and a center annular ring **252**. As shown in FIG. **12**, flexible portions **254** of the sidewalls of the front housing **132** fit within the annular recess **250** when the rear housing **90** is fitted within the front housing **132**. In this manner, the front housing **132** is rotatable relative to the rear housing **90**, but the front and rear housings are axially fixed with respect to each other.

[0084] The rear housing **90** also forms a rear annular recess or notch **253** positioned between the center annular ring **252** and a rear shoulder **256**. Both the rear annular recess **253** and the rear shoulder **256** define a generally square cross-sectional profile.

[0085] The rear housings **90**, as noted above, are received by the yoke **96** through the side slots **98** defined by the yoke **96**. The side slots **98** define C-shaped front cut-outs **257** at the front of the yoke **96**. The portion of the rear housing **90** that includes the square rear annular recess **253** and the rear shoulder **256** are laterally and slidably positioned within the C-shaped cut-outs **257**.

[0086] The internal configuration of the slide slots **98**, including the C-shaped cut-outs **257**, of the yoke **96** and the rear annular recess **253** defined by the rear housing **90**, including the rear shoulder **256**, intermate to rotationally fix the rear housing **90** with respect to the yoke **96**. Thus, when the front housing **132** is rotated with respect to the rear housing **90**, the rear housing **90** and the yoke **96** (and the boot **114**, which is rotationally fixed with respect to the yoke **96**) stay stationary.

[0087] As noted above, the internal tube **84**, which is configured to prevent excess epoxy from leaving the ferrule hub portion **80**, is positioned within both the front housing **132** and the rear housing **90** and extends rearward from the ferrule hub **80**. Since the ferrule hub **80** is rotationally fixed with respect to the front housing **132** and rotates with the front housing **132**, the tube **84** may also rotate with the ferrule hub **80** depending upon the strength of the relative frictional engagement between the tube **84** and the ferrule hub **80**.

[0088] Now referring to FIGS. **7-8** and **21-24**, even though the latch **118** may be designed with a certain amount of elastic flexibility to pivot around a connection point **126**, and the elasticity of the connection point **126** may provide the needed force to return the boot **114** back to a neutral position after the boot **114** has been pulled, in certain example embodiments, the boot **114** may include an additional biasing feature **300** for returning the boot **114**.

[0089] In the example illustrated in FIG. **21**, the biasing feature **300** includes a spring-based return feature **302**. A pocket **140** defined within the yoke **96** may receive a return spring **146**. The pocket **140** of the yoke **96** is configured to position the spring between the rear housings **90**. A rear end **304** of the return spring **146** may be designed to abut a rear wall **306** of the pocket **140** within the yoke **96**, and a front end **308** of the return spring **146** may abut an internal rear face **310** defined within the boot **114** to bias boot **114** toward the forward position of FIG. **7** when released by the user. When the user pulls boot **114** longitudinally away from the connector portions **112**, the spring **146** is compressed. Spring **146** moves the boot **114** back to the rest position of FIG. **7** upon release by the user. The pocket **140** of the yoke **96** may be accessible through an opening such as the aperture **155** defined on the boot **114**.

[0090] Referring now to FIGS. **22-23**, another version of the biasing feature **300** may include integrally molded fingers **312** that are unitarily molded with the yoke **96** that are configured to interact with the boot **114** to return the boot **114** back to a neutral position. In the version of FIGS. **22-23**, the flexible biasing fingers **312** are provided within a horizontal pocket **314** defined on the yoke **96**. The fingers **312** define contact points **316** that can contact an internal rear face **310** defined within the boot **114** to bias the boot **114** toward the forward position. FIG. **23** illustrates the

fingers **312** in a flexed position.

[0091] FIG. **24** illustrates another example of flexible biasing fingers **320** that are integrally molded with the yoke **96**. In the example illustrated in FIG. **24**, the flexible fingers **320** are provided within a vertical pocket **322** positioned between the side slots **98** defined by the yoke **96**. Similar to the example shown in FIGS. **22-23**, the vertical flexible fingers **320** define contact points **324** that can contact the internal rear face **310** defined within the boot **114** to bias the boot **114** toward the forward position.

[0092] By providing the biasing features **300** such as the spring **146** in between the rear housings **90** or by eliminating a separate, removable biasing feature such as a spring and forming the biasing feature as an integral part of the yoke **96** via structures such as the flexible fingers **312**, the overall length of the connector **110** can be reduced.

[0093] FIGS. **25-26** illustrate another example of a duplex connector **410** that has features similar to that of the connector **110** described above. In the example connector **410** shown in FIGS. **25-26**, a yoke portion **496** of the connector **410** is designed for top loading the connector portions **112** instead of a side-loading design such as that illustrated for connector **110** described above.

[0094] As shown in FIG. **25**, the yoke **496** defines side-by-side pockets **457** for receiving the portions of the rear housings **90** that include the square annular recess **253** and the rear shoulder **256** from thereabove. The yoke **496** further defines opposing flexible tabs **411** flanking each pocket **457** that are configured to capture the rear shoulder **256** of the rear housings **90** once the rear housings **90** have been inserted into the pockets **457**. As shown, the internal configuration of the pockets **457** of the yoke **496** and the rear annular recess **253** defined by the rear housing **90**, including the rear shoulder **256**, intermate to rotationally fix the rear housing **90** with respect to the yoke **496**. And, as noted above, the flexible tabs **411** retain the rear housings **90** within the pockets **457** in a vertical direction. Thus, similar to the embodiment of the connector **110** discussed previously, in the connector **410**, when the front housing **132** is rotated with respect to the rear housing **90**, the rear housing **90** and the yoke **496** (and the boot **114**, which is rotationally fixed with respect to the yoke **496**) stay stationary.

[0095] As further shown, a cable path **498** is defined within the yoke **496** for receiving the cables from the top of the yoke **496**. The cable path **498** extends all the way from a crimp region **404** defined by the yoke **496** to the pockets **457**, splitting into two separate sections **489** for leading to each of the pockets **457**. As noted for the embodiment of the connector **110**, the cable path **498** may be designed to guide the fibers directly back into the jacket of the cable and limit deflection or unwanted bending of the fibers if the ferrules at the front of the connector **410** are pressed and moved rearward.

[0096] In the example of the connector **410** shown in FIGS. **25** and **26**, the crimp region **404** of the yoke **496** is similar in configuration to that of the yoke **96** shown in FIG. **15** in that the crimp region **404** is provided as a rear projection **499** that is integrally formed with and extends from the yoke **496**. Similar to the crimp region **104**, the crimp region **404** defines an exterior surface **497** that is textured (e.g., knurled, ridged, provided with small projections, etc.) to assist in retaining the strength members (e.g., aramid yarns) on the yoke **496**. Similar to the crimp region **104** discussed above, in the depicted example of the crimp region **404**, the exterior surface **497** defines a series of parallel rings or ridges **495** extending in a direction from the rear end of the rear projection **499** toward the main body of the yoke **496**. As shown, some of the rings **495** may define discretely spaced scalloped portions **493** that are positioned peripherally around the rings **495**. The scalloped portions **493** define teeth **401** thereinbetween. As noted in the previous examples, the scalloped portions **493** (and the teeth **401**) may be provided in an offset or staggered relationship between adjacent rings **495** as the rings **495** extend front to back. In this manner, when the strength members are pressed underneath a crimp sleeve **106** against the exterior surface **497** of the rear projection **499**, the strength members will tend to spread out and follow zig-zagging patterns as they are captured underneath the crimp sleeve **106**, leading to increased yarn and jacket retention. It is also

contemplated that the teeth **401** may be configured such that they are deformed or smashed against the strength members to provide further retention.

[0097] As shown in the bottom perspective view of the yoke **496** in FIG. **26**, a pocket **403** may be defined by the main body portion of the yoke **496** for accommodating any excess strength members after the strength members have been crimped at the crimp region **404**.

[0098] Referring now to FIG. **27**, another version of a connector **510** that has features similar to connectors **110** and **410** discussed above is illustrated. The version of the connector **510** shown in FIG. **27** has a top-load yoke design similar to the connector **410** shown in FIGS. **25-26**, and includes similar bend or deflection limiting features for the fibers. However, in the connector **510**, the crimp region **504** is provided as a separate crimp body **505** that is removably mounted to a yoke **596** of the connector **510** instead of being integrally formed with the yoke as a rear extension. With the design illustrated in FIG. **27**, the crimp body **505** can be used to crimp the strength members of the cables before the cables are terminated to the connector portions **112**. Once cables have been crimped and terminated to the connector portions **112**, the crimp body **505** can be slidably inserted into a complementary slot **503** defined on the yoke **596**, while the rear shoulders **256** of the rear housings **90** are inserted into pockets **557** defined within the yoke **596**.

[0099] Now referring to FIG. **28**, another example of a connector **610** that has features similar to those of connectors **110**, **410**, and **510** is illustrated, including fiber bend limiting features within the yoke. However, in the connector **610** shown in FIG. **28**, the rear housings and the yoke are combined into a single integral structure **690** forming the rear portion of the connector **610**. Similar to connectors **110**, **410**, and **510** discussed above, the structure **690** defines a front extension **691** that has a front annular ring **651** and a rear annular ring **652** with an annular recess **650** therebetween. As described with respect to the connector **110** discussed above, flexible portions **254** of the sidewalls of the front housings **132** can fit within the annular recess **650** defined by the front extension **691** when each front housing **132** is coupled to the structure **690** of the connector **610**. In this manner, the front housings **132** are rotatable relative to the structure **690** for polarity reversal, but the front housings **132** are axially fixed with respect to structure **690**.

[0100] It should be noted that although the connector **610** is illustrated with a crimp region **604** that is provided as a separate crimp body **605** that is removably mounted to the structure **690**, in other examples, the crimp region can be integrally formed with the structure **690** as a rear extension as discussed for other above examples shown, for example, in FIGS. **15**, **17-19**, **25**, and **26**.

[0101] Referring now to FIG. **29**, another version of a connector **710** that has features similar to connectors **410**, **510**, and **610** discussed above is illustrated. The version of the connector **710** shown in FIG. **29** has a top-load design similar to the connectors **410** and **510** shown in FIGS. **25-27**. However, in the connector **710**, a yoke **796** does not define flexible tabs/latches for retaining the rear shoulders **256** of the rear housings **90** within pockets **757** when the rear housings **90** are inserted into the yoke **796** from above. The rear housings **90** are simply captured against the yoke **796** with a snap-fit cover **705** as shown in FIG. **29**. The cover **705** defines downwardly extending snaps **706** on each of its right and left sides that are configured to be coupled to hooks **707** defined on each side of the yoke **796**. The cover **705** also defines a plurality of tabs **708** adjacent a front end of the cover **705** that fit into snaps **709** defined adjacent the front of the yoke **796** to retain the cover on the yoke **796**.

[0102] As shown, the cover **705** captures against the yoke **796** a biasing member in the form of a spring **700** that is positioned within a center pocket **701** defined on the yoke **796**. The cover **705** defines an aperture **711** for accommodating a portion of the spring **700**. As discussed above, the spring **700** is configured to contact the boot **114** and provide a force to return the boot **114** back to a neutral position after the boot **114** has been pulled.

[0103] It should be noted that the cover **705** may also define protrusions at a bottom side thereof that are configured to keep and constrain the fibers **156**, **158** within fiber pockets **703** defined in the yoke **796** and to limit deflection of the fibers, as discussed above for the previous embodiments.

[0104] Now referring to FIGS. 30-32, another example of a connector **810** having features that are examples of inventive aspects in accordance with the present disclosure is illustrated. The connector **810** is similar to the connector **610** shown in FIG. 28 in the sense that the rear housings and the yoke are molded integrally. However, the combined integral structure is formed from two halves **890**, **892** that are configured to be coupled together in a side-by-side relationship. As will be discussed in further detail below, when the two halves **890**, **892** are brought together, the two halves **890**, **892** capture in between both a biasing member in the form of a spring **800** and strength members of the fiber optic cable that is terminated to the connector **810**.

[0105] As shown in FIGS. 30-32, the two halves **890**, **892** each include intermating hook **893** and snap **894** structures for snap-fitting the two halves **890**, **892** together.

[0106] Slots **895** are provided on each of the halves **890**, **892** to capture rear protrusions **896** of the biasing spring **800** in between the two halves **890**, **892** when the halves are latched together.

[0107] Rear portions of the two halves **890**, **892** are designed such that they capture the strength members of the fiber optic cable in the center, whereas the fibers from the cable are lead around the outside within side slots **898**, similar to the embodiments discussed above.

[0108] In addition to the intermating hook **893** and snap **894** structures, the two halves **890**, **892** also each include interfitting posts **801** and notches **802** for coupling the two halves **890**, **892** together in a side by side relationship.

[0109] Still referring to FIGS. 30-32, as noted above, the rear ends of the two halves **890**, **892** cooperatively define a crimp region **804** of the connector **810**. As shown, similar to the designs discussed above, the crimp region **804**, when formed, may define full annular rings **899** on an exterior surface **897**. However, in contrast to the above-described designs, the crimp region **804** may be designed to capture the strength members in between the two halves **890**, **892** when the halves are brought together. For this purpose, as shown, the two halves each may define an internal textured surface **805** (e.g., knurled, ridged, provided with small projections, teeth, etc.) to assist in retaining the strength members (e.g., aramid yarns) in between the halves. In the depicted embodiment, the two halves **890**, **892** are shown to have inter-fitting teeth **806** that are provided in a relatively offset relationship so as to inter-fit together and capture the strength members of the fiber optic cable in between the two halves **890**, **892**.

[0110] Furthermore, as shown, the halves **890**, **892** may also define a post **809** provided thereinbetween, in front of the crimp region **804**, for wrapping the strength members around and leading them back toward the inter-fitting teeth **806** to provide further strain relief.

[0111] FIG. 32 illustrates the two halves **890**, **892** in an assembled configuration showing the intermating of the various parts.

[0112] FIG. 33 illustrates a top perspective view of a portion of yet another embodiment of a fiber optic connector **910** having features that are examples of inventive aspects in accordance with the present disclosure. The fiber optic connector **910** has features similar to the connector **610** of FIG. 28.

[0113] As shown, similar to the connector **610**, in the connector **910**, the rear housings and the yoke are combined into a single integral structure **990** forming the rear portion of the connector **910**. Similar to connectors **110**, **410**, **510**, and **610** discussed above, the structure **990** defines a front extension **991** that has a front annular ring **951** and a rear annular ring **952** with an annular recess **950** therebetween. As described with respect to the connector **110** discussed above, flexible portions **254** of the sidewalls of the front housings **132** can fit within the annular recess **950** defined by the front extension **991** when each front housing **132** is coupled to the structure **990** of the connector **910**. In this manner, the front housings **132** are rotatable relative to the structure **990** for polarity reversal, but the front housings **132** are axially fixed with respect to structure **990**.

[0114] It should be noted that the connector **910** may define a crimp region **904** that is provided as a separate crimp body that is removably mounted to the structure **990** (such as that shown in FIG. 28), or may define a crimp region that is also integrally formed with the structure **990** as a rear

extension as discussed for other above examples shown, for example, in FIGS. 15, 17-19, 25, and 26.

[0115] The version of the connector **910** requires pre-insertion of the fibers into the structure **990** prior to termination of the fibers to the front housings **132**. And, as shown, a biasing member in the form of a spring **900** is positioned within a center pocket **901** defined by the structure **990**, where the spring **900** is configured to contact the boot **114** and provide a force to return the boot **114** back to a neutral position after the boot **114** has been pulled, as discussed for previous embodiments. The version of the connector **910**, similar to the connector **610**, also requires duplex polishing during termination.

[0116] In the present disclosure, the duplex connectors **110**, **410**, **510**, **610**, **710**, **810**, and **910** are depicted and described as polarity-switching type connectors. Even though the depicted duplex connectors **110**, **410**, **510**, **610**, **710**, **810**, and **910** may include polarity-switching features, as noted above, the inventive latching and unlatching features and other inventive features such as the above-described crimp regions with staggered, scalloped ridges may be used on any duplex or simplex connector, with or without polarity-switching features.

[0117] Although in the foregoing description, terms such as “top,” “bottom,” “front,” “back,” “rear,” “right,” “left,” “upper,” “lower,” “forward,” and “rearward” may have been used for ease of description and illustration, no restriction is intended by such use of the terms. The connectors described herein can be used in any orientation, depending upon the desired application.

[0118] The above specification, examples and data provide a description of the inventive aspects of the disclosure. Many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects of the disclosure.

Claims

1. A telecommunications connector comprising: at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion such that an optical fiber terminated to the ferrule is rotatable with the front housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion.

2-6. (canceled)

7. A telecommunications connector according to claim 1, wherein the at least one connector portion includes a latch for latching the at least one connector portion to a fiber optic adapter, wherein the at least one connector portion includes a boot that is movable longitudinally relative to the rest of the at least one connector portion for interacting with the latch for de-latching the at least one connector portion from the fiber optic adapter.

8-12. (canceled)

13. A telecommunications connector according to claim 7, wherein the at least one connector portion is configured to be latched to an LC format fiber optic adapter.

14. A telecommunications connector according to claim 7, wherein the latch is pivotally connected to the at least one connector portion at a connection portion, wherein the connection portion is an intermediate connection portion positioned between a front end and a rear end of the latch.

15. A telecommunications connector according to claim 7, wherein the front housing portion and

the boot define intermating structures to act as positive stops to stop rotation of the front housing portion with respect to the rear housing portion.

16. A telecommunications connector according to claim 15, wherein the intermating structures are respectively defined by the latch of the front housing portion and a slot on the boot.

17. A telecommunications connector according to claim 1, comprising: two of the at least one connector portions provided in a side-by-side relationship, each connector portion including a front housing portion coupled to a rear housing portion, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions, each front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion.

18. A telecommunications connector according to claim 17, wherein the telecommunications connector is a duplex fiber optic connector and each ferrule terminates a separate optical fiber.

19-33. (canceled)

34. A telecommunications connector comprising: a front housing portion coupled to a rear housing portion and a crimp region, wherein the front housing portion includes a ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion; the crimp region configured for crimping at least a portion of a cable terminated by the ferrule, wherein the crimp region defines a plurality of spaced apart crimp rings stacked longitudinally, each crimp ring defining discretely spaced scalloped portions that are positioned peripherally around the crimp rings, wherein the scalloped portions between adjacent crimp rings are provided in a staggered position in a longitudinal direction defined by the telecommunications connector.

35. A telecommunications connector according to claim 34, wherein the scalloped portions define teeth therebetween that are configured to be deformed against strength members of a cable to be terminated by the ferrule.

36. A telecommunications connector according to claim 34, wherein the telecommunications connector is a fiber optic connector and the ferrule terminates an optical fiber.

37. A telecommunications connector according to claim 34, further comprising a cable terminated by the ferrule crimped to the crimp region.

38. A telecommunications connector according to claim 34, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion.

39. A telecommunications connector according to claim 38, further comprising a yoke coupled to the rear housing portion, wherein the rear housing portion is rotationally and axially fixed with respect to the yoke, the crimp region of the telecommunications connector provided on the yoke.

40. A telecommunications connector according to claim 39, wherein the crimp region is integrally formed with the yoke.

41. A telecommunications connector according to claim 39, further comprising a boot mounted to the yoke, the boot movable longitudinally relative to the yoke, wherein the boot is configured to engage the rear end of the latch and cause the front end of the latch to pivot toward the ferrule of the front housing portion as the boot is moved away from the yoke.

42. (canceled)

43. A telecommunications connector comprising: at least one connector portion including a front housing portion coupled to a rear housing portion, wherein the front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion, the front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that

holds the ferrule, wherein the at least one connector portion includes a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the at least one connector portion at a connection portion, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion; and a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to interact with the latch and cause the latch to pivot relative to the rest of the at least one connector portion as the boot is moved relative to the rear housing portion.

44. A telecommunications connector comprising: two connector portions provided in a side-by-side relationship, each connector portion including a front housing portion, wherein the front housing portions are coupled to a rear housing portion that is formed from two rear housing halves snap-fit together, wherein each front housing portion is rotatable relative to the rear housing portion about a longitudinal axis defined by the front housing portion to change the polarity of the two connector portions, each front housing portion including a ferrule assembly defined by a ferrule and a ferrule hub that holds the ferrule and a latch, wherein the latch defines a front end and a rear end, and is pivotable relative to the rest of the front housing portion at a connection portion, wherein the ferrule assembly is mounted to the front housing portion such that the ferrule hub and the ferrule held by the ferrule hub are rotationally fixed with respect to the front housing portion and rotate together with the front housing portion relative to the rear housing portion when the front housing portion is rotated relative to the rear housing portion, wherein the ferrule hub defines flats on an exterior surface thereof that mate with mating surfaces of a hub mount defined within an interior of the front housing portion for rotationally fixing the ferrule hub and the ferrule with respect to the front housing portion; and a boot mounted on the telecommunications connector, the boot movable longitudinally relative to the rear housing portion, wherein the boot is configured to engage at least a portion of the latch of each of the front housing portions and cause the latch to pivot relative to the rest of the front housing portions as the boot is moved relative to the rear housing portion.

45. A telecommunications connector according to claim 44, further comprising a crimp region cooperatively defined by the two rear housing halves, the crimp region configured for crimping at least a portion of a cable terminated by the telecommunications connector, wherein the crimp region defined by the two rear housing halves is configured such that the two rear housing halves capture strength members of the cable thereinbetween when the two rear housing halves are brought together.

46. A telecommunication connector according to claim 45, wherein each of the two rear housing halves defines surface texturing for capturing the strength members of the cable in between the two rear housing halves when the two rear housing halves are brought together.

47. A telecommunications connector according to claim 46, wherein the surface texturing is provided in the form of intermeshing teeth.

48. A telecommunications connector according to claim 45, further comprising a post positioned between the two rear housing halves for wrapping strength members therearound before capturing the strength members between the two rear housing halves when the two rear housing halves are brought together.
